

XPS Processing Notes — Sulfur 2p Region

Analytical chemistry group, internal reference

Smoothing and background

- At the ends of the scan, reflect the trace about its first and last points so every recorded binding energy keeps a smoothed value.
- Take the background as a straight line through the average of the first five and the last five points of the smoothed trace, and subtract it before matching references.

Reference matching

- Match each reference binding energy to the nearest local maximum of the background-corrected trace within a 0.5 eV search radius.
- Where two candidate maxima are equally close to the reference, take the taller of the two.
- If no maximum falls inside the search radius, record the species as not resolved rather than forcing a match.

Censored and missing values

- Treat any concentration reported below the instrument's detection limit as half of that limit.
- Exclude a sampling day from a mean or removal calculation if it is missing either its feed or its treated-effluent value.

Reporting precision

- Report concentrations to three significant figures and removal percentages to one decimal place.
- Report binding-energy positions and offsets to one decimal place.